PARC T6.2.4 – Methodological Case Study: Frequentist & Bayesian Meta Analysis on Lead Exposure & IQ Loss

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Introduction

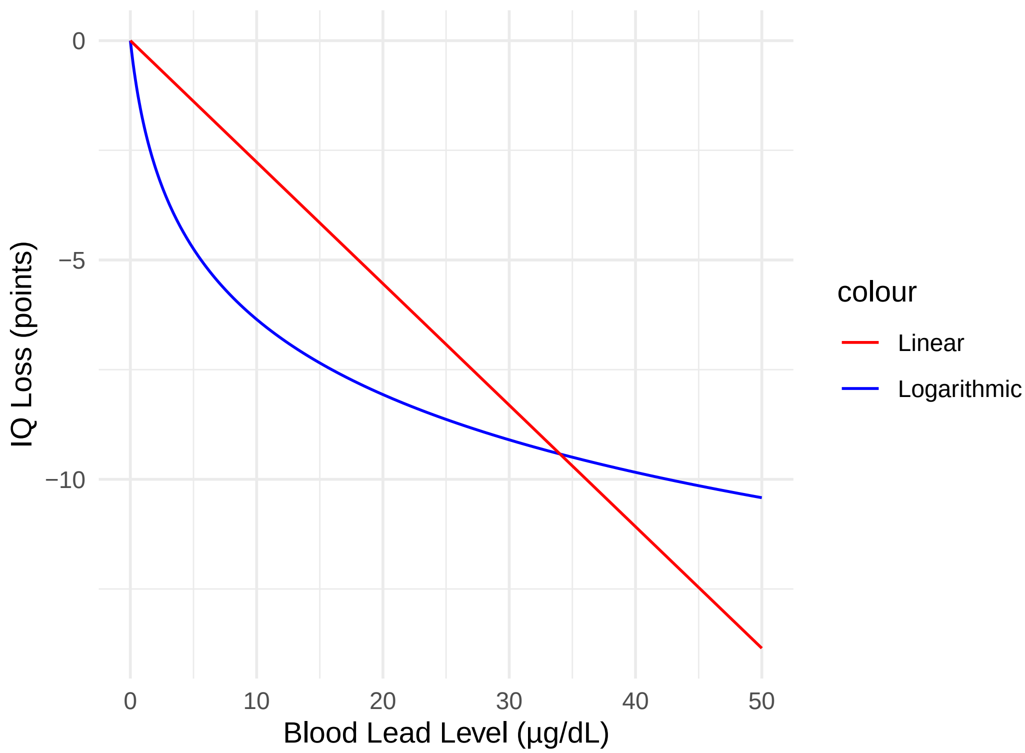
This document describes the methodology of a frequentist and a Bayesian meta analysis on the effects of blood lead levels (BLL) in children (postnatal, concurrent) and their intellectual development (IQ). The results of these meta analyses will be used in Burden of Disease estimation within this methodological case study. The literature review of epidemiological studies was conducted by ANSES.

Harmonizing Effect Estimates

In order to pool the identified effect estimates, we need to transform some of them. Most of the studies transformed their blood lead level (BLL) data to a logarithmic scale in order to fulfill the assumption of a linear relationship between BLL (exposure) and IQ points (outcome) in their regression analyses. They did not always use the same log base. Some used $\log_{10}$, some ln ($\log_e$), and some used $\log_2$. Also, some of the included studies did not transform their BLL at all, so it is on a linear scale. ANSES standardized all estimates to linear increments of BLL, using the template published by Rodriguez Barranco 2017. The following visual tests will explain the rationale behind our decision to transform all effect estimates to a logarithmic scale.

Visualizing Transformations

Comparison of original beta coefficient from Crump et al. 2013 (beta = -2.65, blue) for ln(BLL) to the transformed one (beta = -0.277, red) (now for linear BLL).



As visible in the graph, the effects of lower BLL would likely be underestimated in lower BLL ranges and overestimated in higher BLL ranges. Since today, BLL are often found to be very low (range in GerES V: approximately 0.04 to 4.5 µg/dL), the population health effects (Burden of Disease) of BLL would likely be underestimated. Therefore, effect estimates were transformed to ln(BLL) when necessary in order to harmonize them.

Formulas used to transform effect estimates to ln(BLL):

$$\beta_{ln} = \beta_{linear} \times C_{ref}$$

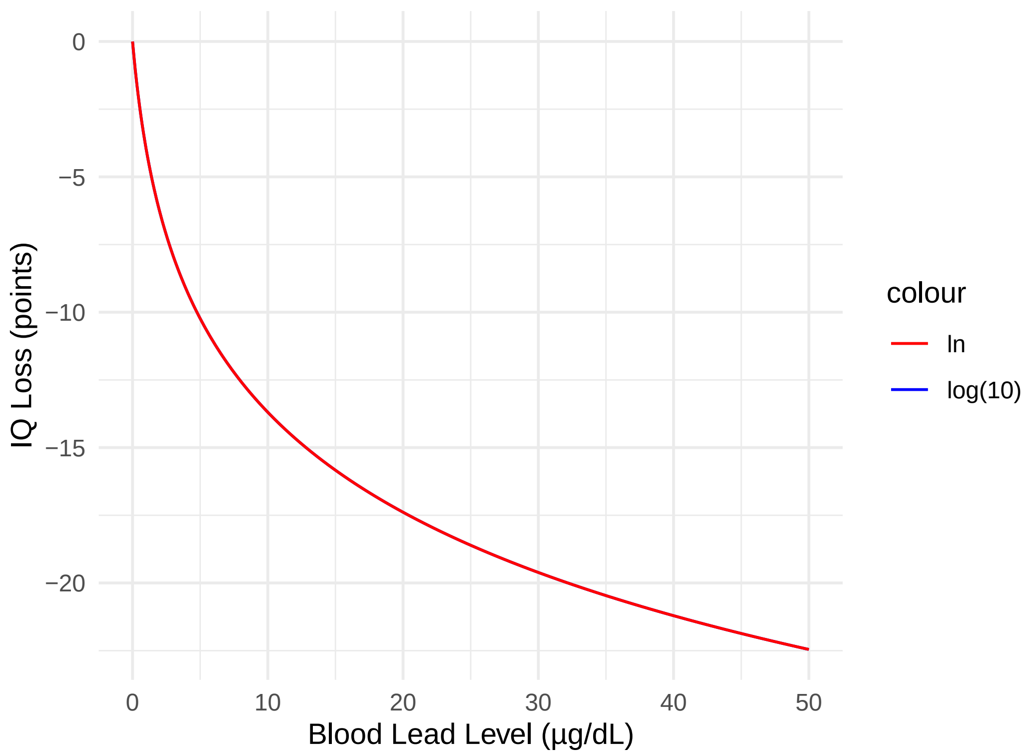
$$\beta_{ln} = \frac{beta_{log_2}}{\ln(2)}$$

$$\beta_{ln} = \frac{beta_{log_{10}}}{\ln(10)}$$

Where:
β_{ln} = harmonized beta coefficient (natural logarithm)
β_{linear} = beta coefficient for linear BLL
β_{log_2} = beta coefficient for log -transformed BLL
$\beta_{log_{10}}$ = beta coefficient for log -transformed BLL
C_{ref} = reference concentration for linear transformation

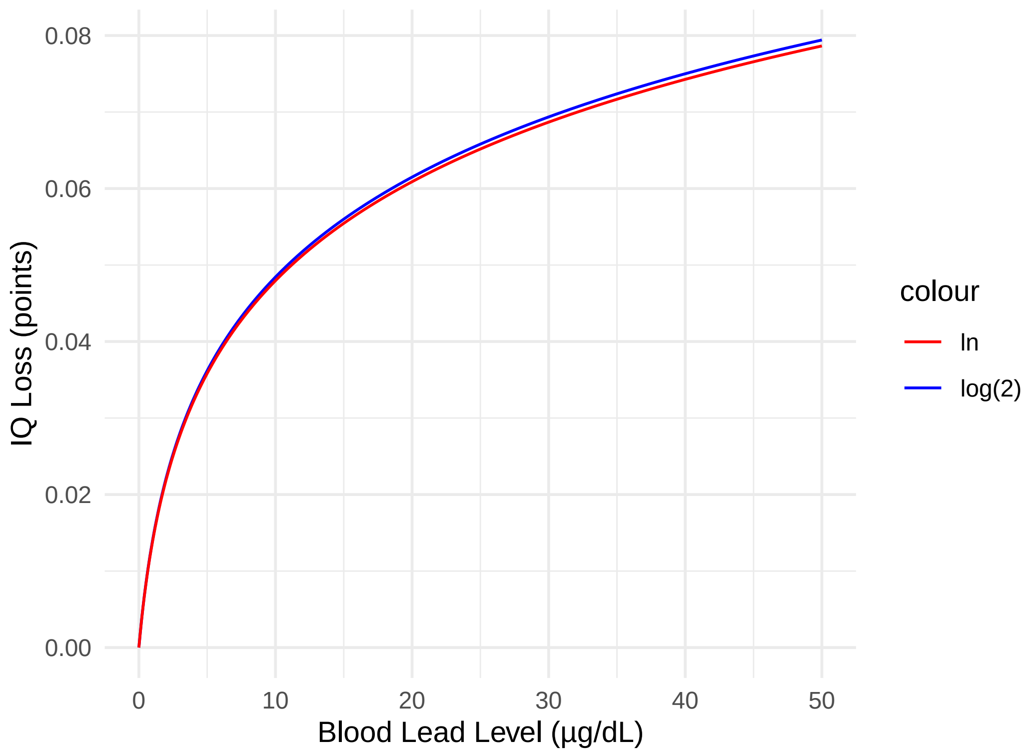
In the following section, all three cases of transformations are visualized in order to check for validity.

Case 1: $\log_{10}$(BLL) to ln(BLL): Dantzer et al. 2020



The transformation from $\log_{10}$(BLL) to ln(BLL) does not affect the exposure-response curve.

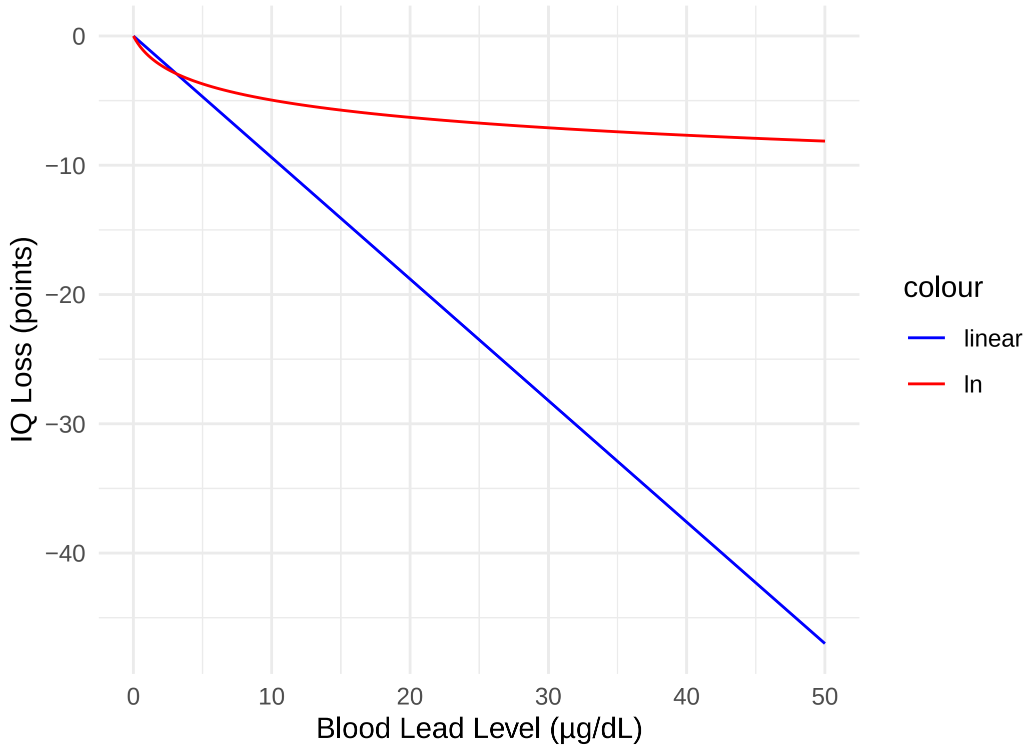
Case 2: $\log_2$(BLL) to ln(BLL): Desrochers-Couture et al. 2018



The transformation from $\log_2$(BLL) to ln(BLL) does not affect the exposure-response curve in the lower exposure ranges. Note: this study reported a small protective effect of lead exposure, this is why the curve is flipped.

Case 3: linear(BLL) to ln(BLL): Iglesias et al. 2011

In order to transform this estimate, we need to set a reference BLL (C_{ref}). 2.2 µg/dL was set in this transformation.



It appears that the transformed estimate does not accurately approximate the original estimate. For this meta analysis, we therefore decided to exclude all studies with estimates for untransformed BLL data.

These are the final studies and transformed effect estimates:

Author, Year	N	Beta	SE(Beta)	Original BLL Scale	Incl. Analysis 1	Incl. Analysis 2	Comment
Bauer 2020	635	-0.1	0.8674	ln	X	X	
Cao 2010							ERF for linear BLL and RCT study design -> excluded
Chen 2005							ERF for linear BLL and RCT study design -> excluded
Crump 2013	1493	-2.65	0.5306	ln	X	X	Overlap with Hornung and Lanphear - Including only this study (Lanphear: linear effect estimate, Hornung: less recent study than Crump)
Dantzer 2020	224	-5.711	1.6109	log10	X	X	
Desrochers-Couture 2018	606	0.0200	0.0622	log2	X	X	

Author, Year	N	Beta	SE(Beta)	Original BLL Scale	Incl. Analysis 1	Incl. Analysis 2	Comment
Earl 2016	127	-5.863	2.215	log10		X	First excluded because 20 month time lag between BLL and IQ measurement
Halabicky 2023				linear			ERF for linear BLL -> excluded
Hong 2015	839	-3.1399	1.365	log10	X	X	
Hornung 2009							Overlap with Crump and Lanphear -> excluded
Iglesias 2011				linear			ERF for linear BLL -> excluded
Lanphear 2019							Overlap with Crump and Hornung -> excluded
Lee 2021	502	-3.83	2.25	ln	X	X	
Lucchini 2012	299	-0.11	0.0842	ln	X	X	
Lucchini 2019	299	-3.483	1.481	ln	X	X	
Menezes-Filho 2018	219	-3.7388	1.2877	log10	X	X	
Min 2009				linear			ERF for linear BLL -> excluded
Roy 2013	708	-4.7	1.658	ln		X	Excluded at first because of unplausibly high effect estimate; first author confirmed missing decimal point in publication
Schnaas 2006	122	0.21	0.8521	ln	X	X	
Wang 2022				linear			ERF for linear BLL -> excluded
Total					10	12	

Results

Analysis 1.1: Bayesian

Setting weakly informative priors:

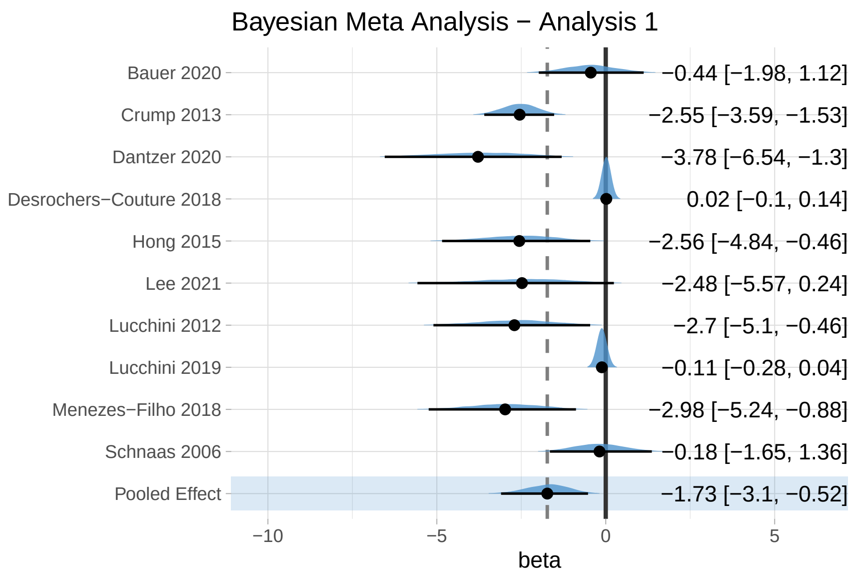
```r
library(brms)

priors <- c(prior(normal(-1, 2), class = Intercept),
            prior(normal(0, 2), class = sd, lb = 0))
```

Fitting Bayesian multilevel random effects model using brms:

```r
library(brms)

m.brm_analysis1 <- brm(
  beta_ln|se(se_beta_ln) ~ 1 + (1|author_year),
  data = data_anaylsis1,
  prior = priors,
  save_pars = save_pars(all = TRUE),
  chains = 4,
  iter = 4000)
```

```
 Family: gaussian
  Links: mu = identity; sigma = identity
Formula: beta_ln | se(se_beta_ln) ~ 1 + (1 | author_year)
   Data: data_anaylsis1 (Number of observations: 10)
  Draws: 4 chains, each with iter = 4000; warmup = 2000; thin = 1;
         total post-warmup draws = 8000

Multilevel Hyperparameters:
~author_year (Number of levels: 10)
              Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
sd(Intercept)     1.80      0.56     0.91     3.09 1.00     2333     3938

Regression Coefficients:
          Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
Intercept    -1.73      0.65    -3.10    -0.52 1.00     1909     3117

Further Distributional Parameters:
      Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
sigma     0.00      0.00     0.00     0.00   NA       NA       NA

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
and Tail_ESS are effective sample size measures, and Rhat is the potential
scale reduction factor on split chains (at convergence, Rhat = 1).
```

Forest plot:



Analysis 1.2: Frequentist

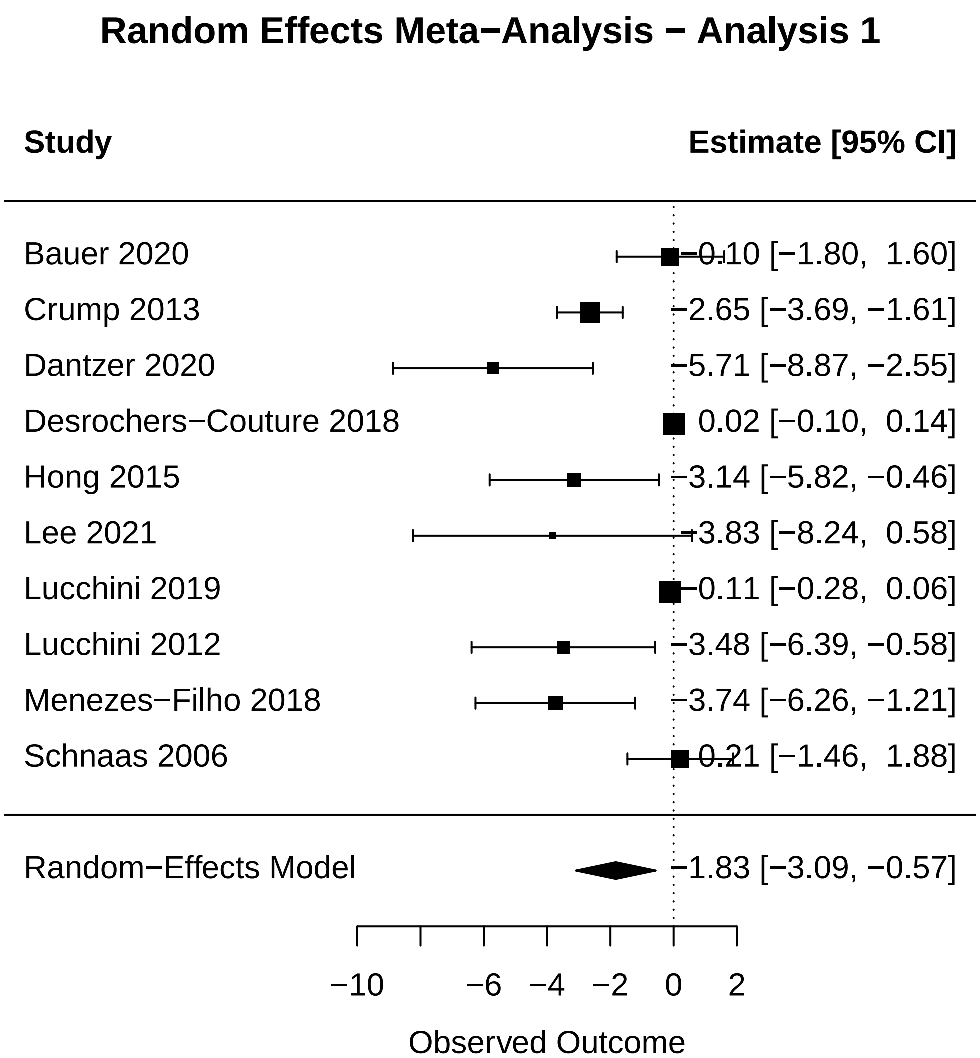
Fitting frequentist random effects meta analysis using metafor:

```
library(metafor)

rma_model1 <- rma(yi = beta_ln,
                  sei = se_beta_ln,
                  data = data_anaylsis1,
                  method = "REML")
```

```
Random-Effects Model (k = 10; tau^2 estimator: REML)

  logLik  deviance       AIC       BIC      AICc
-19.4091   38.8182   42.8182   43.2126   44.8182

tau^2 (estimated amount of total heterogeneity): 3.0078 (SE = 1.9033)
tau (square root of estimated tau^2 value):      1.7343
I^2 (total heterogeneity / total variability):   98.49%
H^2 (total variability / sampling variability):  66.30

Test for Heterogeneity:
Q(df = 9) = 59.6216, p-val < .0001

Model Results:

estimate      se     zval    pval    ci.lb    ci.ub
 -1.8277  0.6434  -2.8407  0.0045  -3.0887  -0.5666  **
```

```
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Forest plot:



Analysis 2.1: Bayesian

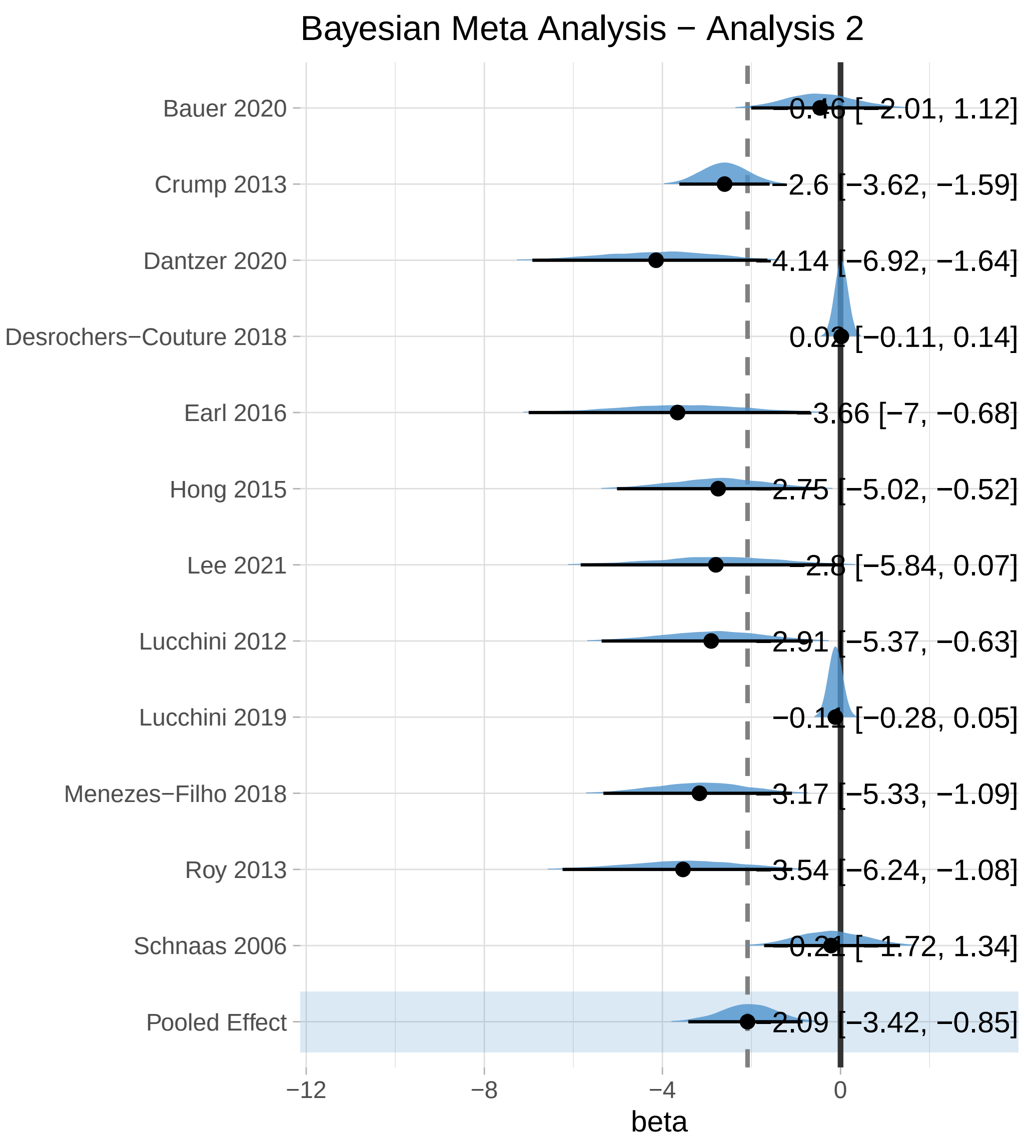
Now including Earl 2016 and Roy 2013.

Fitting Bayesian multilevel random effects model using brms and the same priors as in analysis 1.1:

```r
library(brms)

m.brm_analysis2 <- brm(
  beta_ln|se(se_beta_ln) ~ 1 + (1|author_year),
  data = data_anaylsis2,
  prior = priors,
  save_pars = save_pars(all = TRUE),
  chains = 4,
  iter = 4000)
```

```
 Family: gaussian
  Links: mu = identity; sigma = identity
Formula: beta_ln | se(se_beta_ln) ~ 1 + (1 | author_year)
   Data: data_anaylsis2 (Number of observations: 12)
  Draws: 4 chains, each with iter = 4000; warmup = 2000; thin = 1;
         total post-warmup draws = 8000

Multilevel Hyperparameters:
~author_year (Number of levels: 12)
              Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
sd(Intercept)     1.94      0.55     1.07     3.24 1.00     2342     3409

Regression Coefficients:
          Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
Intercept    -2.09      0.65    -3.42    -0.85 1.00     2005     3130

Further Distributional Parameters:
      Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
sigma     0.00      0.00     0.00     0.00   NA       NA       NA

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
and Tail_ESS are effective sample size measures, and Rhat is the potential
scale reduction factor on split chains (at convergence, Rhat = 1).
```

Forest plot:



Analysis 2.2: Frequentist

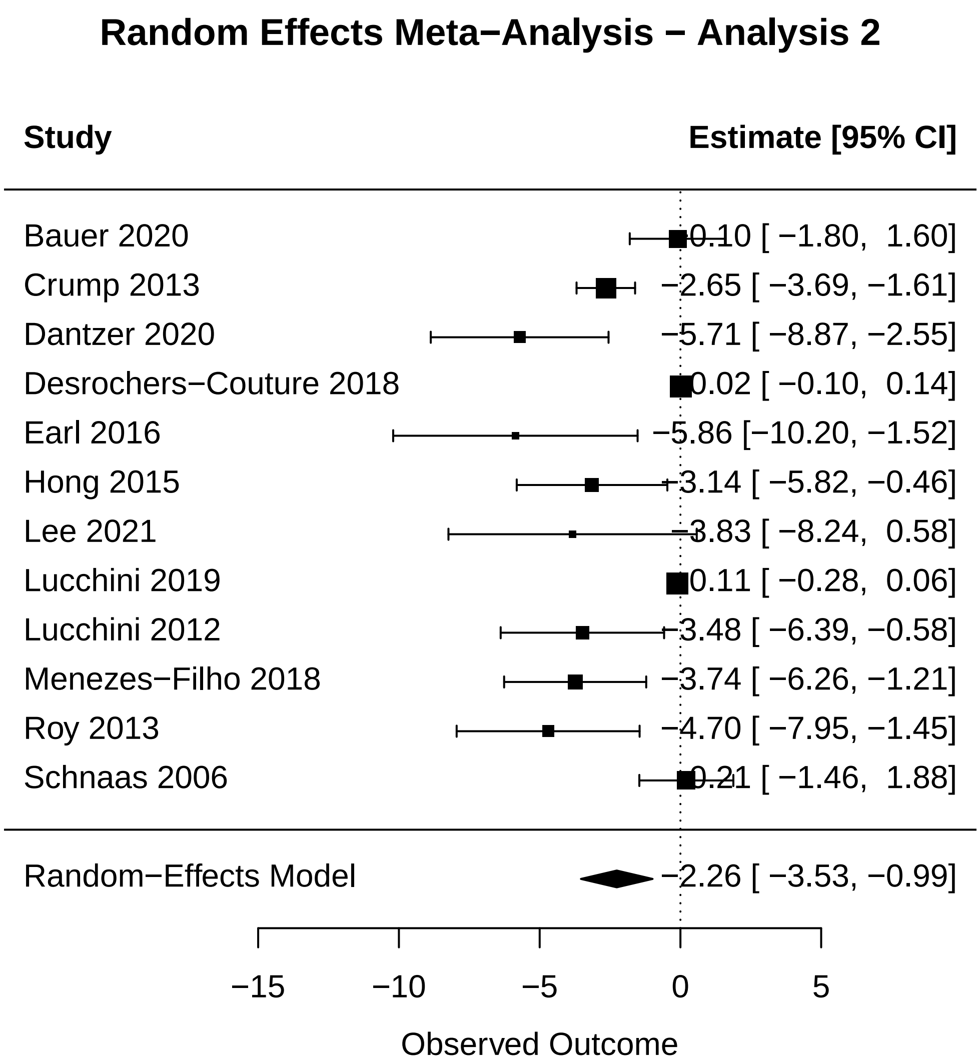
Fitting frequentist random effects meta analysis using metafor:

```
library(metafor)

rma_model2 <- rma(yi = beta_ln,
                  sei = se_beta_ln,
                  data = data_anaylsis2,
                  method = "REML")
```

```
Random-Effects Model (k = 12; tau^2 estimator: REML)
```

```
  logLik  deviance       AIC       BIC      AICc
-24.6351   49.2702   53.2702   54.0660   54.7702

tau^2 (estimated amount of total heterogeneity): 3.6003 (SE = 2.1009)
tau (square root of estimated tau^2 value):      1.8974
I^2 (total heterogeneity / total variability):   98.47%
H^2 (total variability / sampling variability):  65.23

Test for Heterogeneity:
Q(df = 11) = 74.2470, p-val < .0001

Model Results:

estimate      se     zval    pval    ci.lb    ci.ub
 -2.2617  0.6495  -3.4826  0.0005  -3.5346  -0.9888  ***

---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Forest plot:



Comparing results

Comparing the four results:

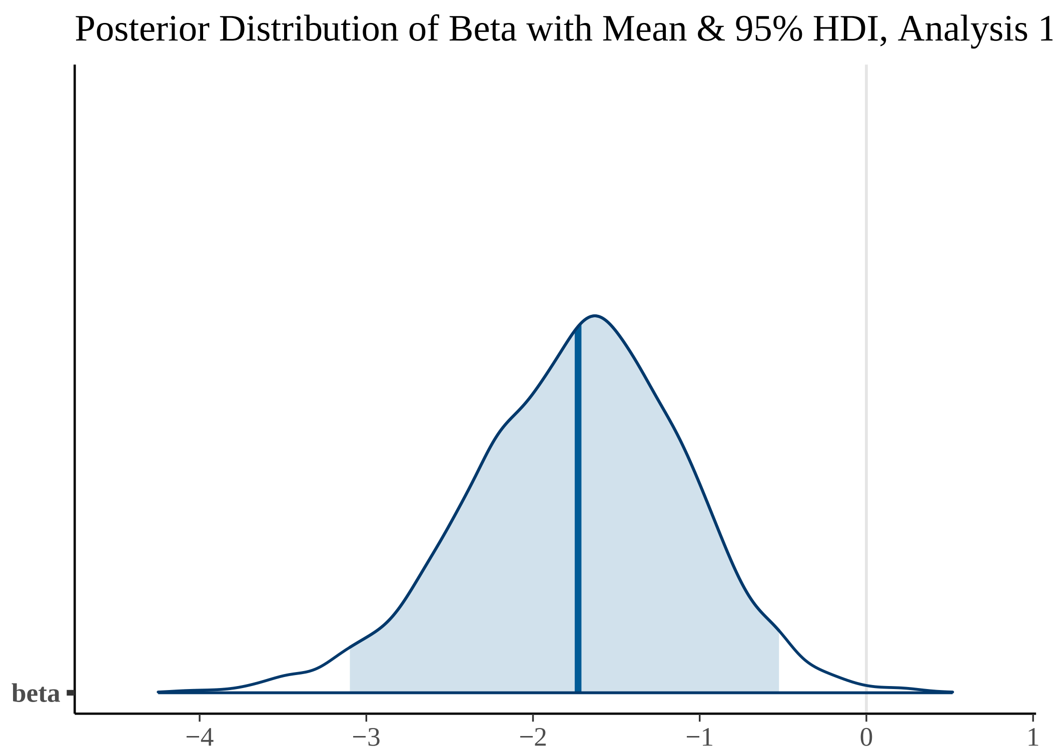
Model	Beta	SE
Bayesian, n = 10 included studies (Analysis 1)	-1.73	0.65
Frequentist, n = 10 included studies (Analysis 1)	-1.83	0.64
Bayesian, n = 12 included studies (Analysis 2)	-2.09	0.65
Frequentist, n = 12 included studies (Analysis 2)	-2.26	0.65

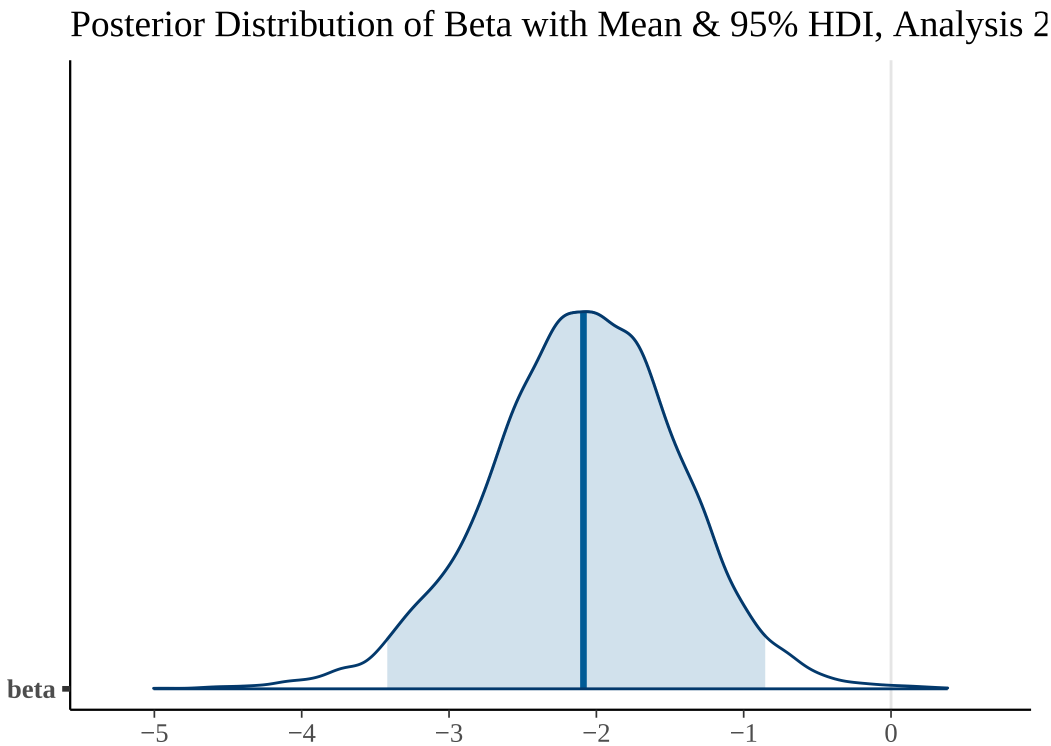
In both analyses, the mean pooled effect resulting from the Bayesian Meta Analysis is smaller than from the frequentist approach. This might be due to a process called shrinkage: The model estimates 'true' effect sizes for each study and if it detects outliers, they will be pulled towards the overall effect. This is an advantage of the Bayesian approach, particularly when pooling a small number of studies. The model is less sensitive to outliers than frequentist models.

I^2 in both frequentist models suggested high heterogeneity (98.49% and 98.47%). In Bayesian statistics, we directly model in-between study heterogeneity. This is why we set a prior for heterogeneity before we ran the model. These are the posterior mean and sd for the heterogeneity of the two models:

Model	Heterogeneity, tau	SE
Bayesian, n = 10 included studies (Analysis 1)	1.8	0.56
Bayesian, n = 12 included studies (Analysis 2)	1.94	0.55

One advantage of the Bayesian approach is that the full posterior distribution of the pooled effect can be used directly in EBD assessments via Monte Carlo simulation. These are the posterior distributions of beta in both analyses:




Posterior Distribution of Beta with Mean & 95% HDI, Analysis 2
beta
−5
−4
−3
−2
−1
0